Dyashin
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Data Science Case Study

Dyashin Technosoft
Jp nagar, Bangalore

Case Study: Project Resource Demand Forecasting System (Dyashin Internal)

AI Case Study – Dyashin Technosoft Pvt Ltd

Company Overview

Dyashin Technosoft Pvt Ltd is an AI-driven innovation startup providing software development, IT consulting, training, and digital solutions across multiple domains. The company handles multiple client projects simultaneously, requiring efficient planning and resource allocation.

Domain

IT Resource Management & Workforce Analytics

Project Duration

1 Month

Tools Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Business Scenario

Dyashin manages multiple projects across domains such as web development, AI solutions, and enterprise applications. Each project requires a dynamic allocation of resources including developers, testers, and analysts.

Currently, resource allocation is mostly based on **manual estimation and past experience**, which leads to:

- Imbalance in workload distribution
- Difficulty in planning upcoming projects
- Inefficient utilization of available workforce

To address this, Dyashin aims to develop a **Project Resource Demand Forecasting System** that uses historical project data to predict future resource requirements.

Instead of reacting to demand, the company wants to **anticipate future needs** and plan resources proactively.

Business Objectives

- Forecast the number of resources required for upcoming weeks/months
- Identify patterns in resource usage across projects
- Improve workforce planning and scheduling
- Support decision-making for project allocation
- Enhance overall operational efficiency

Expected Deliverables

• Python Notebook implementing the predictive model
• Null values will be handled using imputation techniques (mean/mode/median) instead of dropping
• Duplicate records will be analysed and handled appropriately, not blindly removed
• Inconsistent categorical values will be standardized
• Incorrect data types will be corrected
• Outliers will be treated using statistical methods
• Visualization report showing customer segments and engagement patterns
• Model evaluation report including key performance metrics

Expected Outcome

Enhances resource planning by enabling predictive workforce demand forecasting and optimized project allocation.